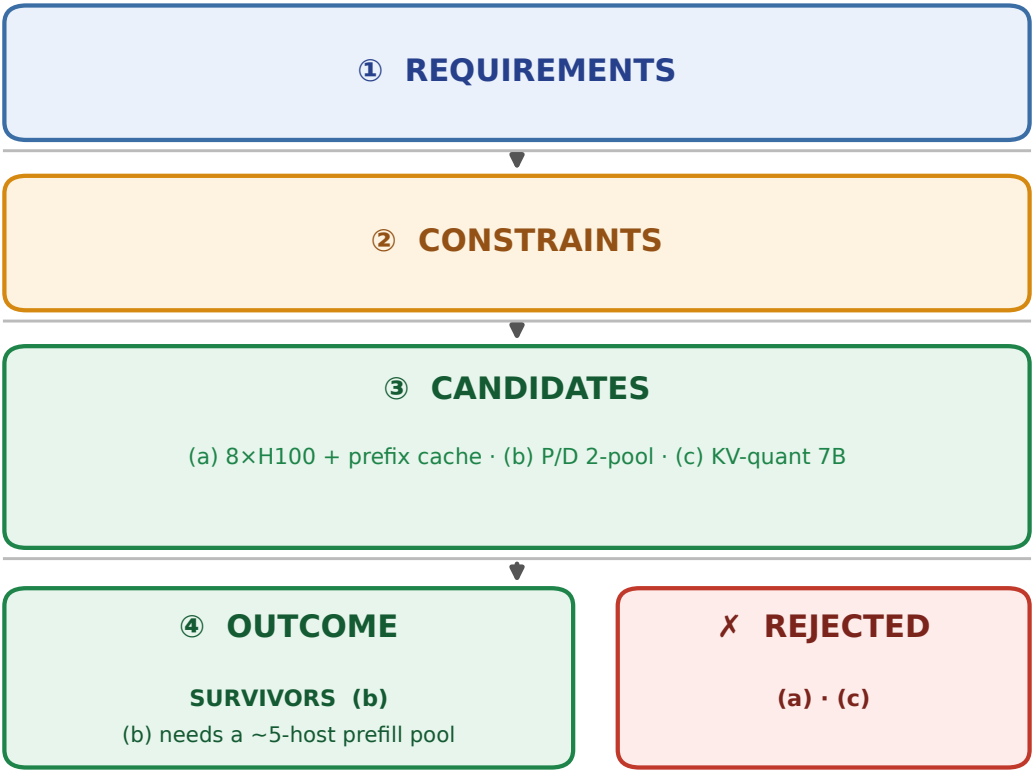Candidate architecture synthesis